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Structure of the Ionosphere and Atmosphere of Saturn From Pioneer 11 Saturn Radio Occultation

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During the flyby of Saturn by Pioneer 11 on September 1, 1979, radio occultation measurements of the ionosphere and upper neutral atmosphere were made near the terminator at latitudes 9.7° south and 11.6° south. The principal electron density peak of the ionosphere occurs at an altitude of about 1,800 km and has a magnitude of 11,400 cm^{-3}, with a sharp lower peak of about 9,000 cm^{-3} at about 1,200 km. The scale height above the main peak corresponds to an exosphere temperature of about 1150 K for an H^+ ionosphere. Ionization appears to extend to 30,000 km, with a broad peak of magnitude 7,000 cm^{-3} at an altitude of about 14,500 km, corresponding to the inner edge of the C ring. The low density of the lower portion of the ionosphere can probably be explained by ring shadowing and equatorial anomaly, and the relatively high densities about 30,000 km, if real, could be explained if densities of $\sim 10^5$ exist at higher latitudes that are connected to the site of measurement by field lines and by plasmaspheric heating by the rings. In the neutral atmosphere, measurements were made to a pressure level of about 180 mbar, showing a temperature inversion region with a triple minimum. The temperature at the principal minimum at 60,344 km is 88 ± 4 K at a pressure of 74 mbar, and there is a prominent minimum about 80 km above the principal one. Comparisons with the temperature structure derived from the infrared radiometer data (Ingersoll et al., this issue) suggest that the helium fraction in the Saturnian atmosphere is 10 ± 4%.

INTRODUCTION

Five years after successfully completing its mission to Jupiter, the Pioneer 11 spacecraft flew by Saturn on September 1, 1979, and conducted a very successful and multifaceted exploration of Saturn and its environment. This was remarkable because the spacecraft was only designed to last 5 years and to communicate scientific data only over earth-Jupiter distances. That it was able to function successfully at Saturn is a tribute to the spacecraft designers of the Pioneer project at NASA Ames Research Center and NASA/JPL (Jet Propulsion Laboratory) Deep Space Network, which made improvements in the ground receiving station radio systems that were more than sufficient to overcome the increased communication distance.

A close flyby of Saturn necessarily brings the spacecraft into occultation by the planet and therefore provides an opportunity to probe its ionosphere and atmosphere with the signal from the spacecraft's communications link. The purpose of our experiment was to learn as much as possible about the structure of the electron density in the ionosphere and the temperature and pressure in the top layers of the neutral atmosphere within the limitations imposed by Pioneer's radio system and by the earth-Saturn-sun geometry at the time of encounter. The Pioneer 11 radio system employs a single *S* band (2.293 GHz) radio link for communication with the earth. Since only one frequency is available, the effects of charged particles cannot be isolated by means of the differential Doppler technique, and so the frequency fluctuations caused by the ionosphere must be laboriously separated from the effects of oscillator drift, drifts introduced by orbital uncertainty, and the effects of interplanetary plasma fluctuations, which at the time of encounter were quite formidable because of the proximity of the earth-Saturn line to the sun. Spacecraft solar conjunction occurred only 10 days after encounter, and on encounter day, Saturn was only 8° from the sun. In the probing of the neutral atmosphere the penetration depth was severely limited by the fact that only about 18 dB of carrier margin were available to offset the effects of defocusing and absorption in the Saturn atmosphere.

In spite of these limitations we feel that the experiment has provided valuable first results on the structure of the ionosphere and neutral atmosphere and on the radius of Saturn, which are discussed in detail below.

The geometry of the Saturn flyby is shown in Figure 1. Figure 1*a* shows the view of Saturn, its rings, and the spacecraft trajectory as seen from the earth. (The oblateness of Saturn, amounting to 0.096 at the 100-mbar level, is not shown in this figure). After flying under the rings of Saturn the spacecraft went behind the limb of the planet at a latitude of about 11.6°S, at 1632:02.4 ET (ephemeris time as observed at the spacecraft). It emerged from behind Saturn at a latitude of about 9.7°S at a time of 1550:47.2 ET. Figure 1*b* shows the view from the trajectory north pole, indicating that the spacecraft was less than 1 Saturn radius from the limb when it entered into occultation at a distance of about 54,100 km. However, at the time of exit this distance had grown to about

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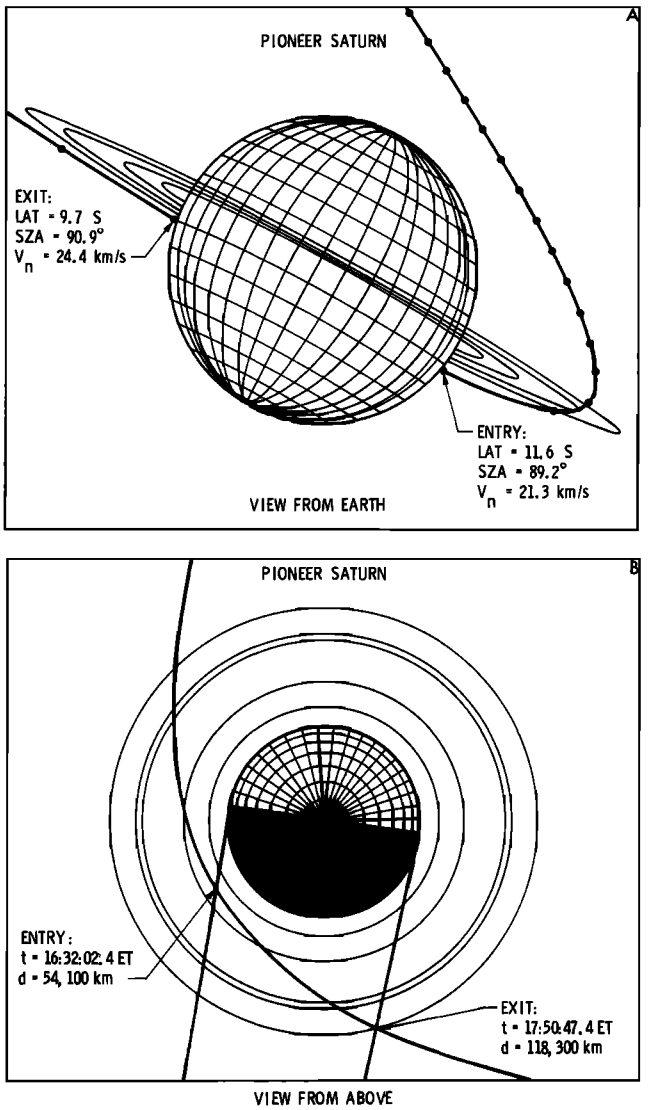


Fig. 1. (*a*) Saturn, its rings, and the trajectory of Pioneer 11 as viewed from the earth. (*b*) View of the flyby from the trajectory north pole.

118,300 km. Consequently, the Fresnel zone size, which defines the maximal resolution of the structure of the atmosphere, is about 2.6 km for the entry and 3.9 km for the exit. Occurring near solar conjunction, the occultation points necessarily were located near the terminator, with the solar zenith angle (SZA) being 89.2° at entry and 90.9° at exit. The radial velocity of the earth-spacecraft line of sight relative to the center of Saturn was about 21.3 km/s at entry and about 24.4 km/s at exit.

Prior to entry into occultation the spacecraft radio system was in the three-way communications mode, in which Deep Space Station 62 (DSS 62) near Madrid, Spain, was transmitting to the spacecraft, while DSS 14 at Goldstone, California, and DSS 63 near Madrid, Spain, were receiving the signal. This was done in order to conserve the telemetry margin at the large 64-m-diameter antenna stations by allowing them to remain in the listen-only mode. This mode was continued all the way into occultation, with the transmitted frequency rates being changed periodically to compensate for the very large changes in Doppler frequency experienced during the close approach to Saturn. During exit, however, the spacecraft was in the one-way mode, receiving its frequency reference from its onboard auxiliary oscillator. Because this oscillator is not stable enough to provide adequate data for the celestial mechanics experiment [*Anderson et al.*, 1980], a compromise was reached in which about 15 min of data after exit were allotted for the occultation experiment in order to provide a baseline, and then the spacecraft was again established in the three-way mode. This amount of time had been thought to be sufficient to provide a baseline for the measurement of the exit ionosphere and was equivalent to radial distance of about 18,000 km, but, as will be explained in another section, it appears to have been somewhat short.

RADIO SYSTEM

The source of data for this experiment was the Deep Space Network (DSN) tracking complex, which also provides all of the telemetry and tracking data. Two very different types of data were used to obtain the results reported in this paper. The first, called for traditional reasons closed loop data, consists of readings of cycle-crossing counts obtained from the metric data assembly (MDA), which obtains its signals from the phase lock loop receivers, which also receive all of the telemetry and tracking data. The cycle-crossing counts, accurate to 0.01 Hz and sampled at a rate of up to 10 samples/s, provide a very convenient data source for radio science; however, closed loop data are only available when the phase lock loop receivers are in lock with the signal. Closed loop lock is usually lost shortly after the effects of the neutral atmosphere become apparent during entry into occultation and is not regained until a considerable time after exit. For data to be used in the derivation of the properties of the neutral atmosphere and portions of the exit ionosphere, open loop data must be relied on. To provide open loop data for radio science, a rather elaborate system has been constructed by the DSN. A block diagram of this system is found in Figure 2 [*Berman and Ramos*, 1980]. The heart of this system is the occultation data assembly (ODA), which, along with the portion of the receiver-exciter subsystem shown in Figure 2, contains all of the instrumentation necessary for the acquisition of open loop data. The receiver-exciter subsystem contains two open loop receivers for *X* band and for *S* band; however, only one, the *S* band receiver, was utilized for this experiment. The most important feature of the system is the programmed oscillator control assembly (POCA), and the programmed oscillator, by means of which the passband of the open loop receivers can be driven according to a preprogrammed function.

This allows a signal to be recorded in a greatly reduced bandwidth, thus providing considerable savings in the time and cost of subsequent data processing. In practice, the orbit of the spacecraft is used in a prediction program which computes the expected time history of the frequency of the received signal, taking into account refraction in the atmosphere. The output of this program, which is operated by the Pioneer navigation team, is given to the DSN Network Operations and Control Center (NOCC), which uses the Predik program to compute a set of polynomial coefficients describing the expected received frequency function. This is then transmitted by a high-speed data line from JPL to the DSN station in question, where it is stored in the Modcomp 11/25 computer. When the ODA is activated to record occultation data, these polynomials drive the POCA, which in turn controls the programmed oscillator to steer its passband with a pre-

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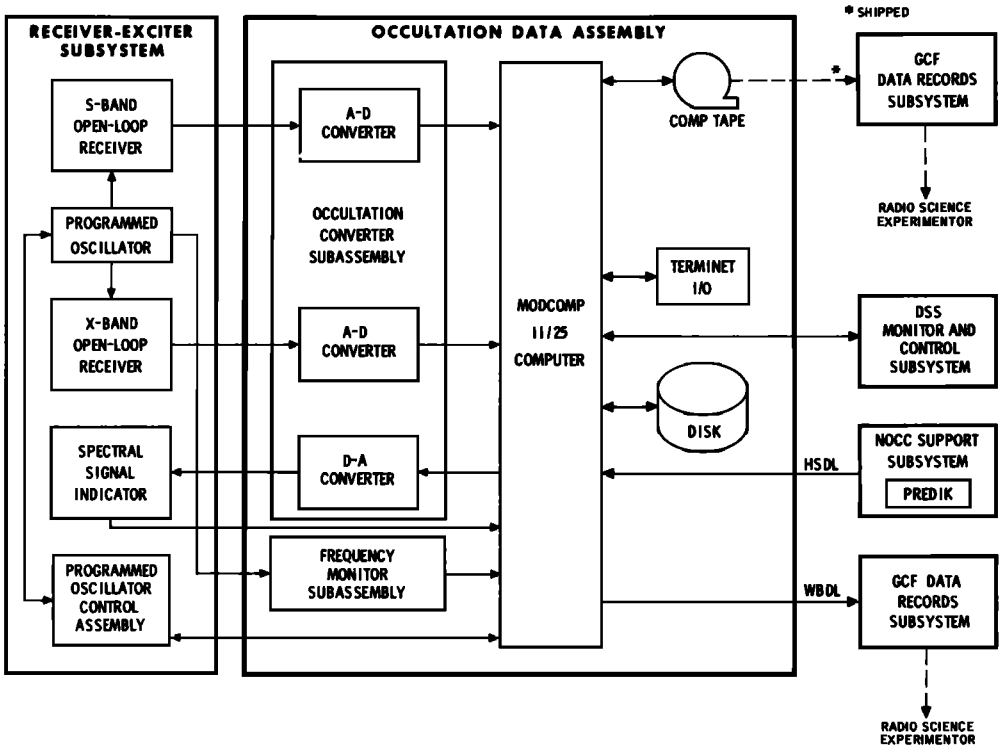


Fig. 2. Block diagram of the occultation data assembly and related DSN equipment which is essential to the acquisition of occultation open loop data.

determined steering function. The bandwidth that was used in the open loop receiver for this experiment was 10 kHz. The signal from the open loop receiver is then passed through an analog-to-digital converter which samples it at the proper rate, in this case 20,000 samples/s. These samples are then processed by the minicomputer and recorded on computer compatible magnetic tape. This tape then forms the basic data record for open loop data. In addition to the sampled signal, this tape also contains information on the function that was used to drive the POCA.

Data Processing and Analysis

In order to obtain the final geophysical quantities that are the objectives of this experiment, the data that were described in the previous section must undergo a large number of processing and analysis steps. The two types of data, open loop and closed loop, are processed in different ways.

In processing closed loop data one does not need to go through the steps of signal detection and processing, which have already been accomplished by the phase lock receiver systems in an analog real-time fashion. The cycle-count data in the intermediate data record (IDR), which are provided for the experimenters by the DSN, are then processed through an IDR stripper program to obtain the received frequency data for the desired DSN stations and time periods. These data are then processed through the radio occultation data analysis program (RODAN), which uses either the spacecraft trajectory information or the predicted received frequencies from the ACCUM link of the JPL navigation double precision orbit determination program (DPODP) to compute frequency residuals, that is, the deviation of the actual received frequency from that which is predicted on the basis of spacecraft orbital motion, rotation of the earth, pole motion, etc. These residuals are then representative of the effects of propagation through the planetary ionosphere and atmosphere, as well as the interplanetary medium between the earth and Saturn.

In the case of the open loop data the processing is far more laborious. The original data record (ODR) tapes that are produced by the ODA system are first processed through a spectrum analysis program which uses a fast Fourier transform algorithm to obtain the frequency of the maximum power point as a function of time. This spectrum file is the basis for the first iteration of signal processing. The signal frequencies obtained from the spectrum are hand edited and extrapolated to provide the input to the programmed local oscillator (PLOSC) program, which computes a local oscillator function. This function is then used in a digital filtering and decimation program [*Woo et al.*, 1973] which uses a digital band-pass filter to filter the data and reduce their bandwidth initially by about a factor of 100 to about 100 Hz. If the local oscillator function were perfect, one would obtain at the output of this program sampled data with a sample rate of 200 samples/s representing a signal stationary in the center of the 100-Hz passband.

The bandwidth-reduced and filtered digital data are then passed through a signal detection program which digitally simulates a second-order phase lock loop receiver and computes the frequency and amplitude of the signal as a function of time. This frequency, which is obtained from the bandwidth-reduced data, is referenced to the center of the passband, and the local oscillator function must be added to it to obtain the true frequency which originally had been residing on the tape obtained from the ODA. In practice, the process of digital filtering and detection must be repeated several times in order to extract a maximal amount of information from the data. In the case of the results described in this paper, great care was taken to insure the maximum accuracy in

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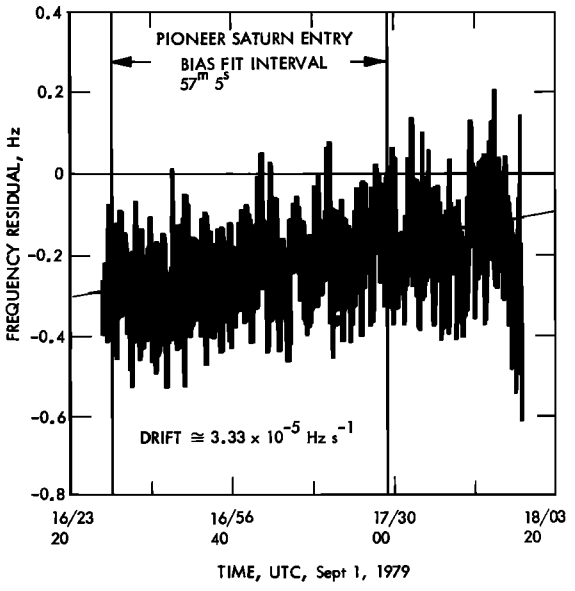


Fig. 3. Frequency residuals obtained from the entry closed loop data covering a time interval of approximately 1.5 hours. An interval of almost 1 hour in extent was chosen to fit the drift function.

the determination of frequency and the maximum depth of penetration into the atmosphere, which require that the rapidly diminishing signal be followed as far into the noise as possible. In order to achieve this goal, spectra of the bandwidth-reduced and decimated signal were computed, and the deviation of the observed signal frequencies from the original local oscillator function was tabulated by hand. The original local oscillator function was then hand edited to incorporate these observed deviations, and the new local oscillator function was used to reprocess the original sampled data through the filtering and decimation program.

The resulting new bandwidth-reduced data were then processed through the phase lock loop receiver program using a bandwidth of 5 Hz. The detection process produced very small deviations from the center frequency up to the point of loss of signal, which was verified by the disappearance of the signal in the spectrum of the decimated data. When the local oscillator was automatically added back in, the output of the detection program produced a precise record of the frequency and amplitude fluctuations of the signal with time. This frequency, however, represented only the frequency observed by the passband of the open loop receiver as it is driven by the programmed oscillator, and it represents the difference between the actual received signal frequency and the steering function provided by the POCA. The POCA steering function was therefore carried along with the other data to that point in the processing. The data are then processed in a manner similar to that described in the discussion of the closed loop data to obtain the frequency residuals which form the basis of subsequent data analysis. In the case of the open loop data, signal amplitudes are also provided. These data will be used to estimate the *S* band absorption coefficient in the region of the atmosphere that was probed by the Pioneer 11 *S* band radio signal.

From this point the analysis of the data is identical for both the open loop and the closed loop data. In the case of the results described in this paper the closed loop residuals were used to produce the ionosphere results, while open loop residuals were used to derive the properties of the lower atmosphere. The frequency residuals that are produced by the RODAN program always show some remnants of bias and drift which are introduced in the case of two-way or three-way data by remaining orbital uncertainties and in the case of one-way data also by oscillator drift. An example of such a case is shown in Figure 3, which shows the frequency residuals for a time interval from approximately 1 hour and 30 min before entry into occultation to a time just prior to entering the top layers of the neutral atmosphere. These were derived from closed loop data with three-point averaging to obtain points every 3 s, or approximately every 64 km in altitude. To correct such data for processing, it is necessary to fit a straight line to a portion of the data which is above the outer limit of the sensible ionosphere and then to subtract all data from this straight line. It is evident from Figure 3 that there is a drift of about 33.3 μHz/s in the residuals, a minute figure but one of great importance in the interpretation of the ionospheric structure. In this case the straight line fit was computed over an interval of 57 min and 5 s, providing approximately the line shown in Figure 3. When the residuals are subtracted from the drift function, the corrected residuals shown in Figure 4*a* are obtained. The primary impression conveyed by Figure 4*a* is that the fluctuations in the frequency on the baseline on the left-hand side, presumably representing solar plasma effects in interplanetary space, are only less by a factor of 2 or 3 than the maximum frequency excursion produced by the main peak of the Saturnian ionosphere (the downward excursion at the far right). This is also borne out in Figure 4*b*, which shows the integrated frequency residuals (equivalent to the phase path difference or change) as a function of time. The quasi-periodic fluctuations to the left of 1730:00 represent effects of the solar wind plasma electrons, while the excursion downward from that point presumably is the effect of the ionosphere of Saturn. Figure 4*b* also shows part of the rationale for picking the bias fit interval shown in Figure 3. By repeated trial and error it was found that intervals terminating to the left of 1730:00 led to phase path delay curves similar to that in Figure 4*b*; however, if the bias fit interval was extended beyond this limit, a distortion occurred, showing a large positive phase excursion. The linearity of the large amount of data used to fit the bias and drift function (almost 1 hour) also indicates the high quality of the spacecraft orbit that was used to compute these residuals.

The corresponding frequency residuals for the exit data, showing the effects of the Saturnian ionosphere, are displayed in Figure 5. These residuals, obtained from open loop data, contain only a span of 13 m and 20 s, which corresponds only to about $\frac{1}{7}$ of the entry data shown in Figures 3 and 4. The approximate drift rate in this case is about 438 μHz/s, about an order of magnitude higher than that for the entry. However, this is mostly due to the drift rate of the auxiliary oscillator of the spacecraft radio system. Because of the very short data span, it was impossible to actually fit a drift function to the exit ionosphere data. However, in order to be able to compare features in the electron density profiles between the entry and exit data, and not for the purpose of comparing the magnitudes, an artificial drift function was produced, which produces an approximate match between the magnitudes of the peak electron density between entry and exit profiles.

After the corrected frequency residuals are produced, they are then used along with a trajectory of the spacecraft to compute the refractive bending angle and the corresponding ray

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asymptote distance for each data point. This procedure has been described previously [*Kliore and Woiceshyn,* 1976] and will not be discussed in detail here. In short, the mass, gravity field harmonics, and rotation rate of Saturn are used to compute an equipotential surface, which then is used to locate a center of refraction corresponding to a normal to the equipotential surface. This is then used as the center in the computation of the refractive bending angle and the ray asymptote distance [*Kliore,* 1972].

The data are then inverted by using the Abel integral transform method [*Fjeldbo and Eshleman,* 1968; *Kliore,* 1972] to obtain the index of refraction as a function of radial distance from the center. These data form the basic product of the radio occultation analysis.

The refractivity can then be used directly to compute the electron density in the ionosphere. In order to compute quantities such as density, pressure, and temperature in the neutral atmosphere it is necessary to specify the composition of the atmosphere. Once the composition is specified, the barometric equation is integrated to obtain the pressure as a function of altitude, and the perfect gas law is then applied to compute the corresponding temperature. For an oblate planet, such as Saturn, the equatorial radius corresponding to each measurement point is computed on the basis of the latitude and a constant oblateness value based on an ellipsoidal approximation.

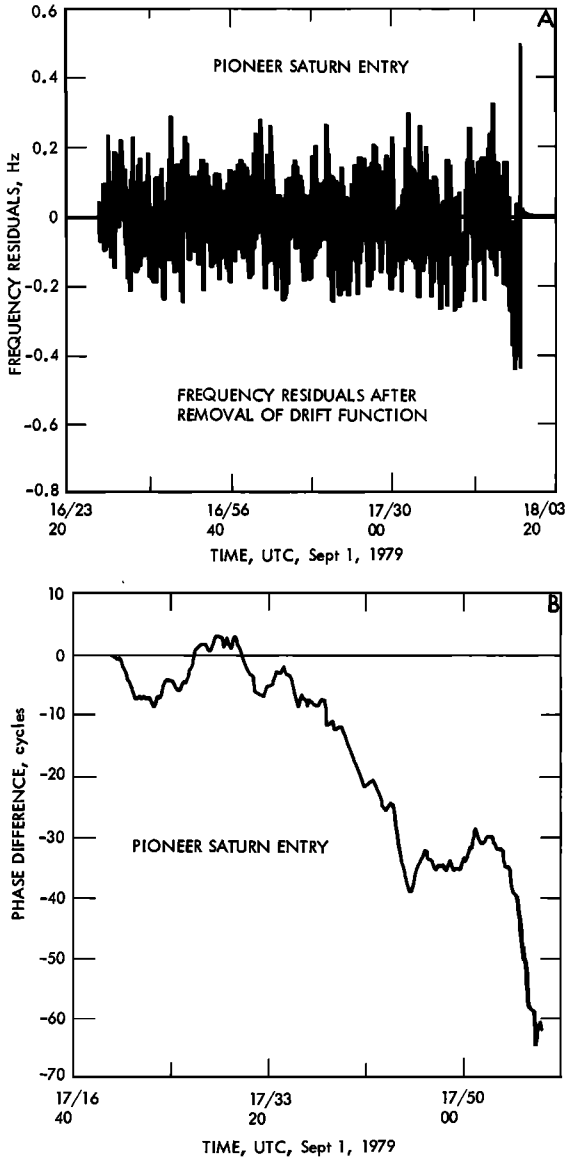


Fig. 4 (*a*) Frequency residuals of Figure 3 after the drift and bias have been removed. Only data to the right of 1730:00 were used to compute the structure of the Saturn ionosphere. (*b*) Phase path changes in *S* band cycles (about 6.5 cm in the two-way mode), computed by integrating with respect to time the frequency residuals shown in Figure 4*a*.

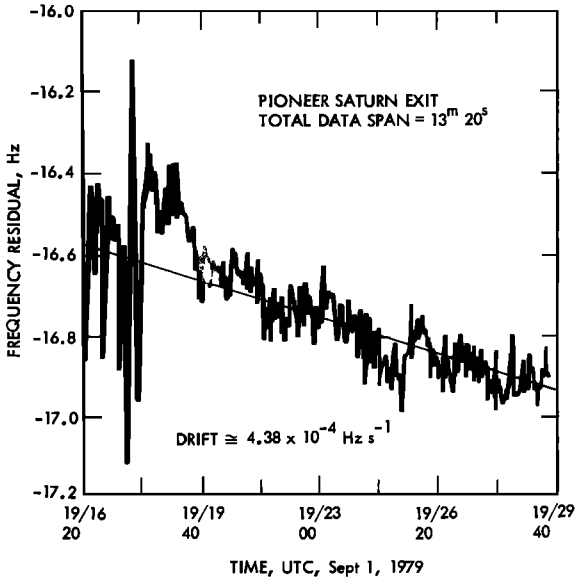


Fig. 5. Frequency residuals from exit open loop data. The time span covered by these data was not sufficient to provide a baseline for the fitting of a drift function.

THE IONOSPHERE

In our preliminary paper describing the structure of the ionosphere of Saturn [*Kliore et al.,* 1980] we mentioned that the published profile may represent only a lower peak of a more extensive ionosphere. To a degree, this seems to have been the case, although conclusions regarding the low electron density at the peak remain virtually unaffected. The original profiles were derived from the approximately 15 min of open loop data for both entry and exit. An intriguing frequency fluctuation near the end of the open loop data encouraged us to review all available closed loop data. The results of this examination appear in Figures 3 and 4 discussed above.

The ionospheric occultation data were affected by interplanetary solar wind electron fluctuations due to the proximity of the communication link to the sun. It is interesting to investigate the effect of these fluctuations in terms of the equivalent spherically stratified electron density. A portion of the baseline data above 90,000 km was processed and inverted to obtain the electron density fluctuation profile shown in Figure 6. It is evident that the equivalent peak to peak electron density variations, which are due to the interplanetary conditions and not associated with Saturn, are of the order of 3×10^3 cm^{-3} peak to peak. This value, along with possible uncertainties in the orbit and drift function, led to the error bars shown in Figure 9.

The electron density profile derived from the entry data is shown as the solid curve in Figures 7*a* and 7*b*. Figure 7*a* shows the profile extending out to an equatorial radius of

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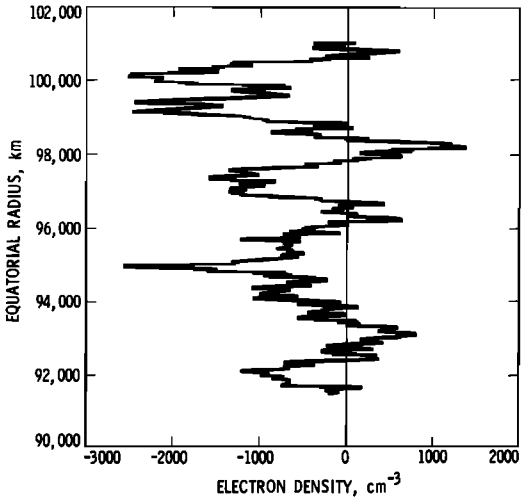


Fig. 6. Electron density variations equivalent to frequency fluctuations produced by solar wind electrons in interplanetary space. These equivalent electron densities were obtained by inverting data within the baseline used for the fitting of the drift function.

70,000 km, and Figure 7*b* continues from 70,000 to 80,000 km. Ionization was observed to a radius of approximately 90,000 km, although, for reasons discussed in the preceding paragraph, the electron densities below 3×10^3 cm^{-3} may very well be spurious. The profiles represented by a dotted line in Figures 7*a* and 7*b* refer to the exit data. It must be stressed again, however, that no quantitative conclusions can be formed from the exit profile, since it was obtained with an artificially generated drift function and hence can only be used for qualitative comparison of features. The structure of the main peaks (Figure 7*a*) for both the exit and the entry data appears to be remarkably similar, with the exception that the upper peak of the entry profile which has a density of about 10^4 cm^{-3} seems to have transformed into a shoulder of about 8.5×10^3 cm^{-3} in the exit profile. The main peak occurs at an altitude of about 1,800 km above the neutral density level of 10^{19} cm^{-3}, with a peak density of about 1.14×10^4 cm^{-3}, and there is a clear suggestion of a sharp lower peak of about 9×10^3 cm^{-3} peak density about 600 km below the main peak. Unfortunately, the full extent of the lower peak is not visible in the entry data because all data below that point were affected by an anomaly which will be further described in a subsequent section. The difference of about 1200 km between the altitude of the main peak in the entry and exit profiles is too large to be explained by trajectory errors or processing uncertainties, and its explanation will have to await further evaluation.

Figure 7*b* shows an interesting feature at about 75,000 km, which appears to be real, although its magnitude, about 5×10^3 cm^{-3}, is less than twice the peak to peak fluctuation introduced by the solar wind plasma. The location of this feature is quite intriguing because it is at the radius corresponding to the inner edge of the C ring [*Gehrels*, this issue], and it is possible that this feature is connected with it in some manner.

Prior to the Pioneer Saturn encounter, *Waite et al.* [1979] attempted to predict the ionospheric structure of Saturn. The nominal model (model III, Figure 9) with an eddy diffusion coefficient of $\sim 10^6$ cm^2 s^{-1} and a high exospheric temperature of 1350 K resulted in an H^+ ionosphere with a corresponding large plasma scale height. The peak electron density of 10^5 cm^{-3} was located at an altitude of 2250 km (above an atmospheric density reference level of 1×10^{19} cm^{-3}). An ionosphere composed of H^+ is consistent with the large vertical extent of the ionospheric measurements made by Pioneer Saturn. In particular, if we fit a slope to the ionospheric data between 2000 and 6000 km (Figure 9), the measured scale height corresponds to an exosphere temperature of ~1150 K (if $T_i = T_e = T_n$) for an H^+ ionosphere, whereas an exospheric temperature of >10,000 K would be indicated for an ionosphere composed of hydrocarbon ions. Even an H_3^+ ionosphere would require an exospheric temperature of ~3450 K to explain the observed scale height. Furthermore, earlier rocket measurements by *Weiser et al.* [1977] and later satellite observations by J. Clarke (private communication, 1980) give a resonantly scattered Lyman α from the Saturn disk that

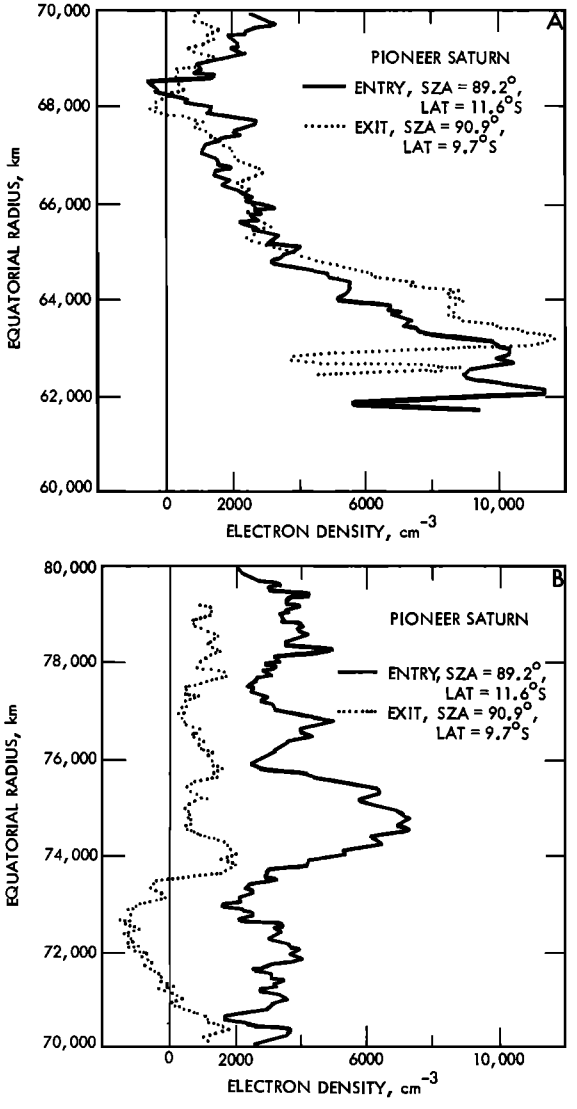


Fig. 7 (*a*) Electron density in the ionosphere of Saturn between equatorial radii of 60,000 and 70,000 km. Solid curve is a profile produced from the closed loop data taken during entry. The dashed curve is a profile obtained from exit data by using an artificial drift function and should be used only for comparison of features and not for magnitudes of electron density. (*b*) Electron density in the ionosphere of Saturn between equatorial radii of 70,000 and 80,000 km. The feature at about 75,000 km corresponds approximately in radial distance to the inner edge of the C ring.

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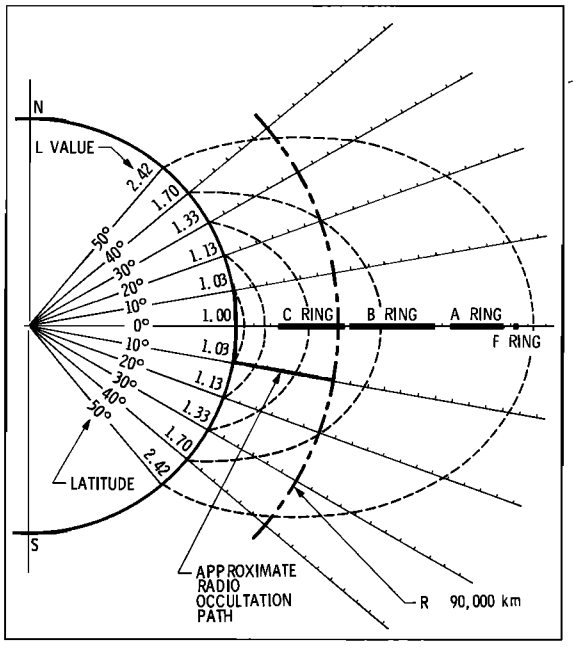


Fig. 8. The path of the radio occultation measurement relative to the Saturn dipole and position of the rings.

would imply an eddy coefficient of $\sim 10^6$ cm^2 s^{-1} [*Wallace and Hunten*, 1973], consistent with our earlier choice of parameters in model III, which resulted in an H^+ ionosphere. The obvious problem with the prediction of *Waite et al.* [1979] is the peak electron density. The predicted peak electron density due to photoionization is 10^5 cm^{-3}, whereas the measurements indicated 10^4 cm^{-3}. This discrepancy implies that model III of Waite et al., would require a hundredfold decrease in ion production to reproduce the measured peak density.

The decrease in the peak electron density of our post-Pioneer model from the earlier model III of *Waite et al.* [1979] (Figure 9) is the result of two effects, the choice of an average solar zenith angle of 80° and a fivefold decrease in the solar flux due to ring shadowing. The choice of an average solar zenith angle of 80° was selected to approximate the solar zenith angle of the measurements. However, on a rapidly rotating planet, where the major ion has a lifetime that is long with respect to the rotation period, the choice of an average solar zenith angle for various latitudes and longitudes needs more detailed consideration. The more important decrease in the peak electron density of the post-Pioneer model is due to ring shadowing. We have used the calculation of *Brinkman and McGregor* [1979], which suggest a fivefold decrease in the solar flux for the current Saturn season at 10°S latitude.

Similar decreases in the plasma density could be the result

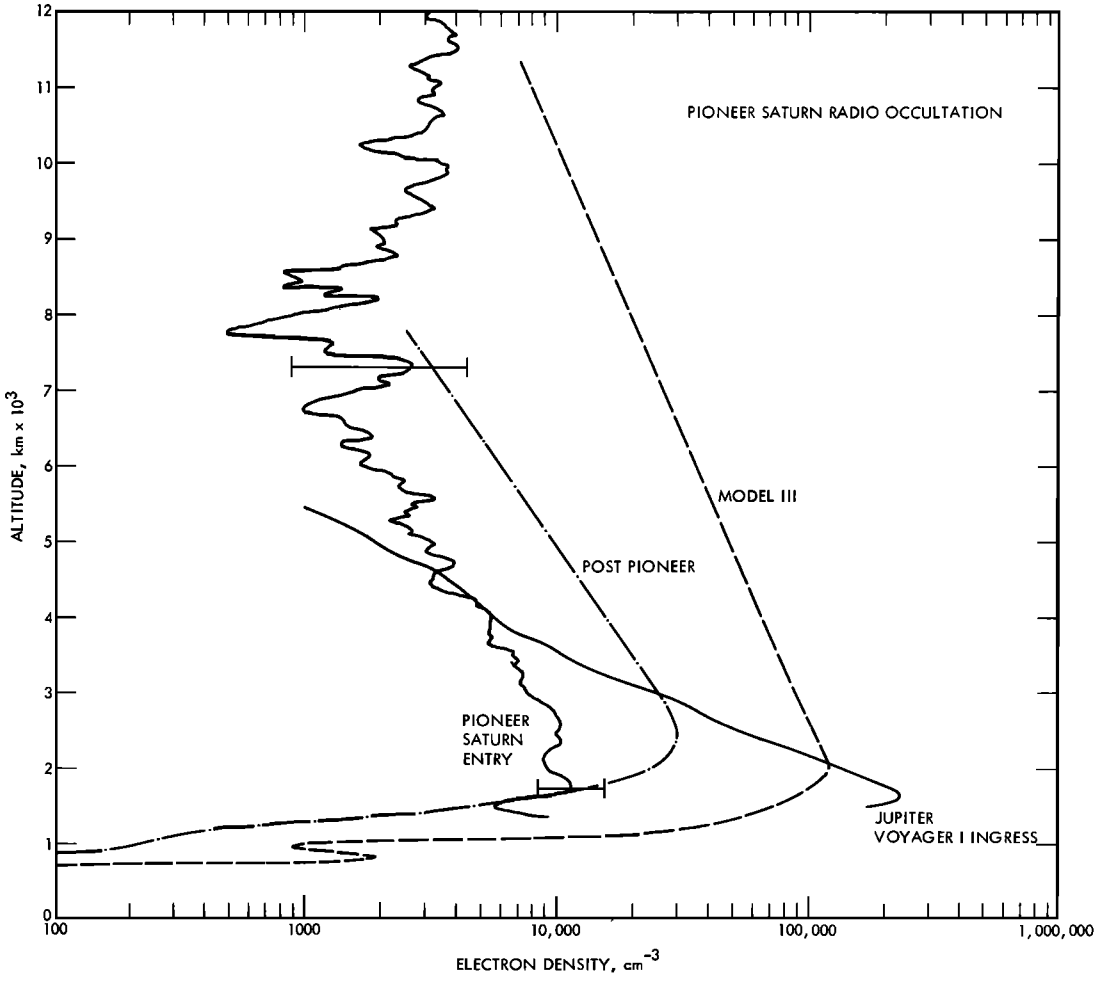


Fig. 9. Electron density in the terminator ionosphere of Saturn compared with the Jupiter ionosphere as measured by Voyager 1 [*Eshleman et al.*, 1979] and two theoretical models described in this paper. The zero-altitude level was chosen at an equatorial radius of 60,328, corresponding to the 10^{19} number density level.

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TABLE 1. Saturn Atmo phere Structure From Exit Measurement (Latitude, 9.5°S; SZA, 90.9°)

Equatorial Radius, km	Refractivity, n-units	Temperature, K	Pressure, mbar	Temperature Uncertainty,* K	Pressure Uncertainty, mbar
60496.4	0.5	121.6	1.8	−11.5, +21.1	−02., +0.4
60490.1	0.6	121.2	2.1		
60483.8	0.7	120.4	2.4		
60477.9	0.8	118.7	2.7		
60472.0	0.9	117.2	3.0		
60466.7	1.0	114.6	3.4	−7.7, +12.6	−0.3, +0.4
60461.6	1.1	111.2	3.7		
60456.8	1.3	107.8	4.1		
60452.4	1.5	104.4	4.6		
60448.1	1.7	101.2	5.1		
60444.2	1.9	98.3	5.6	−4.7, +7.3	−0.4, +0.4
60440.2	2.2	95.0	6.1		
60436.1	2.4	94.4	6.8		
60432.1	2.7	93.5	7.5		
60428.6	3.0	92.8	8.2		
60424.5	3.4	92.7	9.1	−3.1, +4.5	−0.4, +0.5
60420.8	3.7	92.9	10.1		
60417.4	4.0	93.1	11.0		
60413.8	4.4	93.6	12.1		
60410.0	4.8	94.6	13.3		
60406.9	5.1	95.3	14.3	−2.2, +3.1	−0.3, +0.6
60403.9	5.5	95.8	15.5		
60401.4	5.8	96.0	16.5		
60398.9	6.2	96.0	17.1		
60396.9	6.5	95.8	18.4		
60395.0	6.9	95.4	19.3	−1.7, +2.4	−0.4, +0.6
60392.7	7.3	95.1	20.5		
60390.5	7.7	95.0	21.6		
60388.6	8.1	94.8	22.7		
60386.9	8.5	94.5	23.7		
60385.1	8.9	94.2	24.8	−1.4, +1.9	−0.5, +0.5
60383.7	9.3	93.8	25.7		
60382.2	9.7	93.4	26.7		
60380.9	10.1	93.0	27.6		
60379.4	10.5	92.6	28.7		
60378.0	10.9	92.2	29.7	−1.2, +1.5	−0.4, +0.6
60376.7	11.4	91.8	30.8		
60375.3	11.8	91.5	31.9		
60374.0	12.3	91.2	33.1		
60372.6	12.8	90.9	34.2		
60371.4	13.2	90.7	35.4	−1.0, +1.3	−0.4, +0.6
60370.3	13.6	90.5	36.4		
60369.3	14.1	90.2	37.4		
60368.3	14.5	89.9	38.4		
60367.1	15.0	89.7	39.6		
60365.9	15.5	89.5	41.0	−0.9, +1.1	−0.5, +0.6
60364.6	16.1	89.4	42.4		
60363.4	16.6	89.3	43.8		
60362.3	17.1	89.3	45.1		
60361.4	17.5	89.2	46.1		
60360.7	17.9	89.1	47.0	−0.8, +1.0	−0.4, +0.6
60359.8	18.4	88.9	48.2		
60358.9	18.9	88.8	49.4		
60357.9	19.4	88.7	50.7		
60357.1	19.9	88.6	51.9		
60356.2	20.4	88.5	53.1	−0.7, +0.9	−0.5, +0.6
60355.4	20.9	88.4	54.3		
60354.6	21.3	88.3	55.5		
60353.8	21.8	88.2	56.7		
60352.8	22.4	88.1	58.2		
60351.8	23.0	88.1	59.8	−0.6, +0.8	−0.5, +0.6
60350.9	23.6	88.2	61.3		
60350.1	24.1	88.2	62.7		
60349.3	24.6	88.2	64.0		
60348.7	25.1	88.2	65.2		
60347.9	25.6	88.2	66.4	−0.6, +0.7	−0.5, +0.6
60347.4	26.0	88.1	67.4		
60346.7	26.5	88.1	68.7		
60346.0	27.0	88.0	70.0		
60345.3	27.5	88.0	71.3		
60344.8	27.9	88.0	72.4	−0.6, +0.6	−0.5, +0.6
60343.9	28.6	87.9	74.1		
60343.1	29.2	88.0	75.7		

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TABLE 1. (continued)

Equatorial Radius, km	Refractivity, n-units	Temperature, K	Pressure, mbar	Temperature Uncertainty,* K	Pressure Uncertainty, mbar
60342.2	29.9	88.1	77.6		
60341.6	30.4	88.1	79.0		
60340.6	31.1	88.3	81.0	−0.5, +0.6	−0.5, +0.6
60339.9	31.7	88.4	82.7		
60339.1	32.4	88.5	84.4		
60338.4	32.9	88.7	86.0		
60337.5	33.7	88.9	88.3		
60336.9	34.2	89.0	89.6	−0.4, +0.6	−0.5, +0.6
60336.3	34.7	89.1	91.2		
60335.7	35.2	89.2	92.5		
60335.1	35.8	89.3	94.2		
60334.4	36.4	89.4	95.9		
60333.7	37.0	89.6	97.7	−0.4, +0.5	−0.5, +0.6
60333.0	37.6	89.7	99.6		
60332.1	38.4	90.0	101.9		
60331.4	39.1	90.2	103.9		
60330.7	39.7	90.4	105.6		
60329.9	40.4	90.6	107.9	−0.4, +0.5	−0.5, +0.6
60329.5	40.8	90.7	109.1		
60329.0	41.4	90.8	110.8		
60328.4	41.9	91.0	112.3		
60327.8	42.5	91.1	114.3		
60327.1	43.2	91.3	116.2	−0.4, +0.5	−0.5, +0.6
60326.1	44.1	91.6	119.2		
60325.3	45.0	92.0	121.9		
60324.4	45.8	92.4	124.7		
60323.7	46.5	92.7	126.9		
60323.1	47.2	92.9	129.0	−0.3, +0.4	−0.5, +0.6
60322.7	47.5	93.1	130.3		
60322.2	48.1	93.2	132.1		
60321.5	48.9	93.4	134.5		
60321.1	49.3	93.4	135.7		
60320.6	49.7	93.5	137.0	−0.3, +0.4	−0.5, +0.6
60320.3	50.3	93.6	138.6		
60319.8	50.9	93.6	140.3		
60319.2	51.6	63.7	142.5		
60318.7	52.3	93.8	144.4		
60318.4	52.7	93.8	145.6	−0.3, +0.4	−0.5, +0.6
60317.7	53.5	93.9	148.2		
60317.3	54.0	93.9	149.6		
60316.8	54.7	93.9	151.4		
60316.5	55.2	93.9	152.7		
60316.0	55.8	93.9	154.5	−0.3, +0.4	−0.5, +0.6
60315.5	56.6	93.9	156.6		
60315.1	57.2	93.9	158.3		
60314.5	58.0	93.9	160.4		
60313.8	59.0	93.9	163.4		
60313.3	59.8	93.9	165.5	−0.3, +0.3	−0.5, +0.6
60312.8	60.6	93.8	167.6		
60312.4	61.4	93.8	169.6		
60311.8	62.3	93.8	172.2		
60311.1	63.3	93.8	175.0		
60310.7	64.1	93.7	177.0	−0.3, +0.3	−0.6, +0.7
60310.1	65.0	93.6	179.4		
60309.5	66.1	93.5	182.2		

*The temperature uncertainties tabulated here reflect only the contribution of the fluctuations in the refractivity baseline and of the choice of initial temperature and are valid only for a composition of 10% He and 90% H_2. The uncertainty in the composition introduces an additional temperature uncertainty of ±3 K.

of field-aligned outflows of H^+. This outflow could be produced by asymmetric conditions along the 'plasma flux tube' brought about by ring shadowing in the conjugate region. The outflow of H^+ may also be due to other ring-associated conditions in the plasmasphere, such as ion heating or loss of ionization in the ring plane.

Both the outflow of H^+ and the decreased insolation due to direct ring shielding result in peak electron densities of 3×10^4 cm^{-3} and are insufficient to explain the measured density of 1×10^4 cm^{-3}. Since the eddy diffusion coefficient is probably of the order of 10^6 cm^2 s^{-1} and the electron peak is at an altitude of 1800 km, we do not feel that loss of H^+ by reaction with CH_4 can be responsible for this additional depletion. The current knowledge of the upper atmosphere would lead us to believe that the low densities are not the result of chemical loss processes but are most probably due to additional plasma transport processes.

The large time constants involved in the plasma loss processes (>10^5 s) provide a strong case for the influence of plasma transport on the ionospheric profile. The possibility exists that

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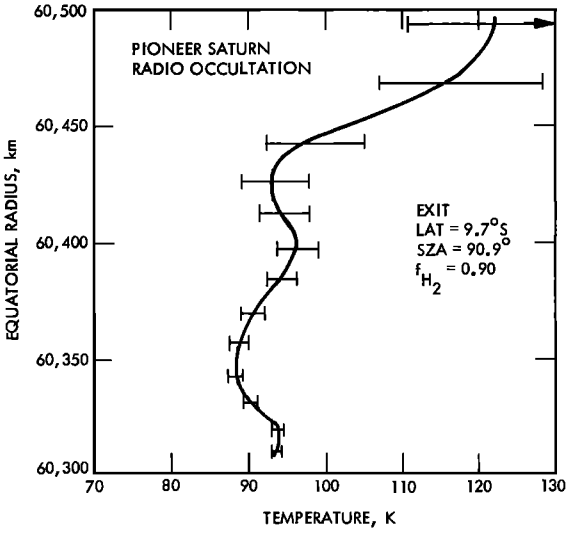


Fig. 10. Temperature profile in the neutral atmosphere of Saturn derived from open loop data taken during exit from occultation. The error bars represent extremes that are obtained by taking into account a fluctuation in the zero-refractivity level at the top of the atmosphere by ±0.02 n-units.

vertical drifts of the Saturn ionosphere occur as a result of neutral winds and electric fields. A 0.2-mV m^{-1} westward electric field resulting in a 10-m s^{-1} vertical drift coupled with a plasma time constant of 10^5 s leads to a 1000-km vertical displacement of the ionosphere [*Rishbeth et al.*, 1978]. Such vertical drifts of the ionosphere could possibly explain the observed difference in the altitude of the peak electron density between the entry and exit profiles, which has been noted earlier in this discussion. The vertical drift of the ionosphere, coupled with the curvature of the near-horizontal magnetic field lines, might lead to a decrease in the equatorial peak density and an increase in density at higher latitudes (the so-called 'fountain effect' observed in the terrestrial equatorial anomaly). A Saturnian equatorial anomaly may help explain the low equatorial ionospheric densities. The possibility of an equatorial anomaly existing on Saturn is currently being investigated.

Perhaps the most intriguing feature of the Saturn ionosphere is its large measurable extent (~30,000 km for $N_e > 10^2$ cm^{-3}), if it is indeed real. Not only does the ionosphere cease to fall off with a diffusive scale height above 68,000 km, it also increases to an apparent secondary maximum at a radial distance of ~74,500 km (see Figure 7*b*). However, the interpretation of the vertical profile is complicated by the near-horizontal orientation of the magnetic field lines and the preponderance of plasma transport along field lines. As a result, the plasma at large radial distances near the equatorial plane connects via field lines to the planetary ionosphere at higher latitudes (see Figure 8). The possibility exists then that what we are seeing in the vertical profile of Figure 9 is mainly due to a latitudinal effect in the lower ionosphere. If the lower equatorial ionosphere is indeed 'anomalous,' then higher-latitude electron densities in the lower ionosphere may be of the order of 10^5 cm^{-3}. In this case, electron density values of 3×10^3 cm^{-3} at 77,500 km (Figure 7*b*) would result from a plasma in near-diffusive equilibrium if the plasma temperature ($T_e + T_i$) were on the average several thousand Kelvin along the 'plasma flux tube' originating at 30°S latitude (see Figure 8).

The fact that the secondary ionospheric maximum corresponds closely with the inner edge of the C ring (Figure 8) appears to be more than a mere coincidence and would seem to imply some type of interaction between the ionosphere and rings.

We see in Figure 8 that for a magnetic dipole field aligned with the planetary spin axis, the magnetic field lines, and thus the 'plasma flux tubes,' are perpendicular to the equatorial plane. Therefore the ring plane will separate the hemispheres of many of the 'plasma flux tubes.' For instance, in the case of the C ring, any flux tubes originating in the planetary ionosphere between 25° and 36° latitude will intersect the C ring plane in the plasmasphere. The other important part of the dynamical picture is the differential azimuthal velocity between the plasmasphere, which presumably corotates with the planet with a 10-hour period, and the ring particles, which follow Keplerian orbits. The implied differential velocity ranges from ~5 km s^{-1} at the outer edge of the C ring to ~10 km s^{-1} at the inner edge of the C ring.

The ring plane cuts through the plasma flux tubes over a wide range of L values (1.22–2.36), impeding or stopping in-

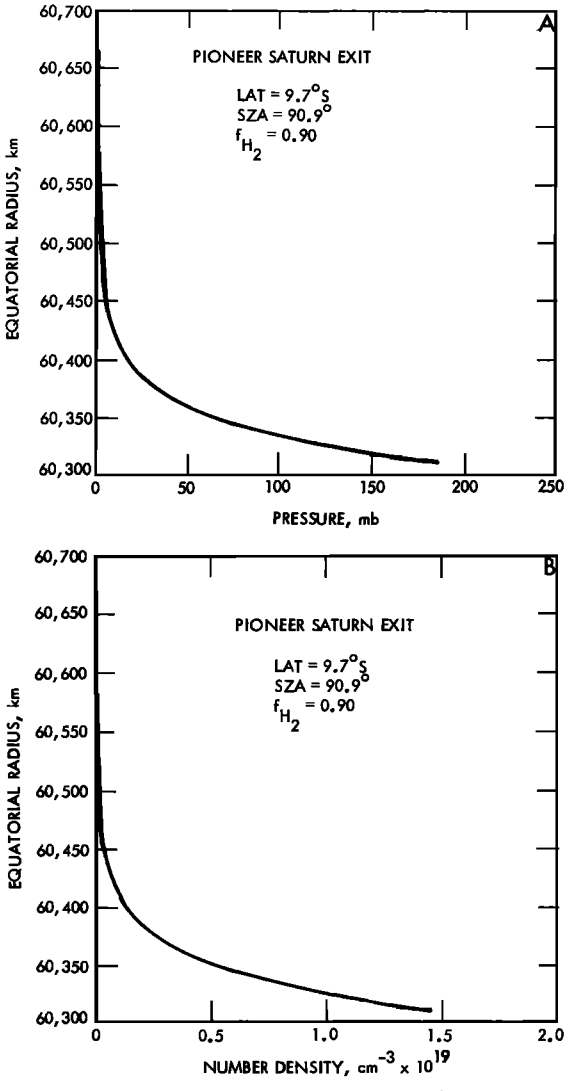


Fig. 11. (*a*) Atmospheric pressure in the atmosphere of Saturn derived from exit occultation data. (*b*) Number density derived from exit occultation data.

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terhemispherical plasma flow depending on the ring thickness and the density of the particles in the rings which interact with H^+. Some of the H^+ will sputter or charge exchange with the ring particles, forming an ionization sink. Other H^+ ions will be scattered by the ring plane, changing their pitch angle and possibly magnetically trapping them near the ring plane. More importantly, these scattered ions will be heated due to the differential rotation velocity between the ring particles and the plasmasphere. The amount of H^+ scattered rather than absorbed will depend strongly on the material of the rings.

We can argue from this discussion that the observed electron densities between 70,000 and 80,000 km may be due to a combination of two factors: (1) The lower equatorial ionosphere measurements are 'anomalously low,' and at higher latitudes (22°S–33°S), where the 70,000- to 80,000-km 'plasma flux tubes' originate, the lower planetary ionosphere electron densities will be closer to the predicted peak electron density of 10^5 cm^{-3} [*Waite et al.*, 1979] and possibly enhanced by a 'fountain effect.' (2) The interaction of the plasmaspheric H^+ with the ring particles or heating in the lower ionosphere by Joule dissipation or particle precipitation will provide sufficient ion, electron, or neutral heat sources to boost the plasma temperatures in the upper ionosphere to the several thousand Kelvin necessary to explain the required plasma scale height of ~7800 km.

If the mid-latitude electron densities in the lower ionosphere turn out to be consistently low (10^4 cm^{-3}), or the heat sources insufficient, more exotic mechanisms may have to be invoked to explain the observed plasmaspheric densities. Possibilities include sputtering of the rings by H^+, followed by photoionization of the neutral fragments. When these newly formed ions are swept up by the corotating magnetic field, some additional ion heating will result. Further ionospheric measurements at higher planetary latitudes must be carried out before we can assess the need for such endogenous ion sources.

Planetary missions have been full of surprises, and the Saturn ionosphere is no exception. Much more information will be needed for us to begin to understand the complex processes operative in the equatorial Saturn ionosphere. Detailed calculations must be performed to explain the anomalously 'low' densities in the lower ionosphere and to verify and explain the surprisingly 'high' densities of the plasmasphere.

THE ATMOSPHERE

Although the open loop data were analyzed to the disappearance of the signal for both the entry and the exit cases, the entry data exhibit anomalous behavior below the level of the ionospheric peak, and for that reason they have not been reliably analyzed. Therefore all results relating to the structure of the upper portions of the neutral atmosphere of Saturn presented in this section are based on the analysis of open loop data obtained during the exit from occultation.

Before the carrier margin of 18 dB was exhausted, the radio beam penetrated the sensible neutral atmosphere of Saturn for a vertical distance of about 300 km. The values of refractivity, temperature, and pressure for approximately the lower 200 km of this range of penetration are listed in Table 1, which covers the range of pressures from 1.85 to 182.2 mbar, corresponding to equatorial radii ranging from 60309.5 to 60496.4 km. The temperatures listed in Table 1 are also plotted against equatorial radius in Figure 10. The error bars shown are not based on variations introduced by the choice of

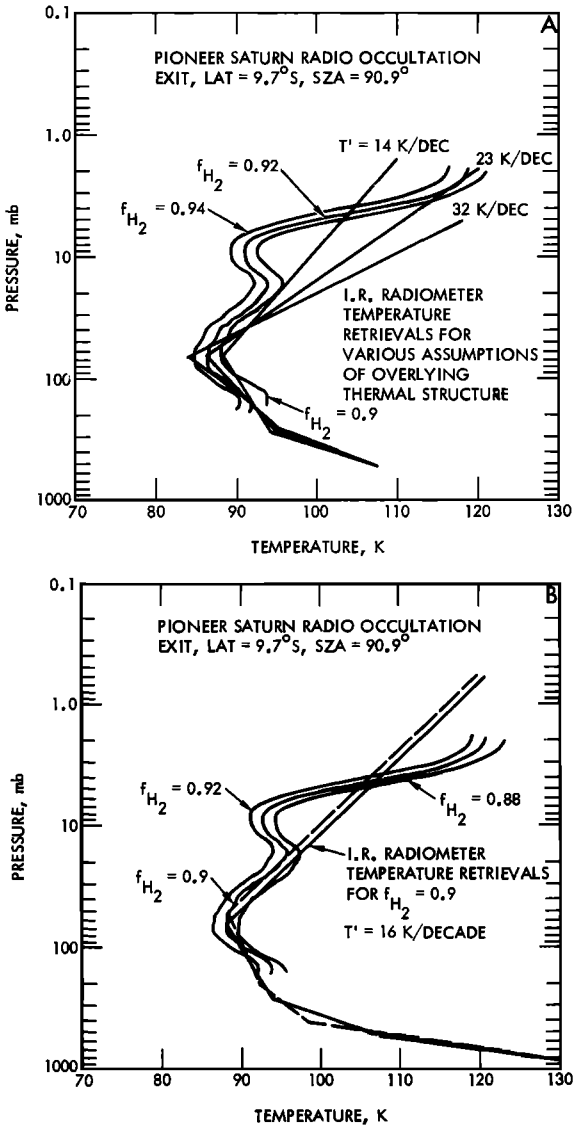


Fig. 12. (*a*) Comparison of infrared radiometer–retrieved temperature structures for various assumptions of the overlying thermal structure and occultation temperature profiles computed for various fractions of hydrogen and helium. (*b*) Comparison of infrared radiometer temperature retrievals for a hydrogen fraction of 90% and a steepness of the overlying structure of 16 K/decade with occultation temperature profiles computed for fractions of 88, 90, and 92% hydrogen. The best correspondence between the two results is obtained for a fraction of helium by number of 10%.

initial temperatures (50–150 K), which have very little effect at the pressure level shown. Rather, they are based on extremes that can be obtained from the data by allowing the level of zero refractivity to vary within the limits of noise fluctuations at the top end of the refractivity data. In the case of the Pioneer Saturn exit data, noise fluctuations in the region between the bottom of the ionosphere and the top of the atmosphere introduce a fluctuation in refractivity with a peak to peak value of about 0.04 n-units. The nominal values of temperature were obtained when the zero-refractivity level was chosen at the most likely location of the commencement of the atmospheric effects, and the limits for the error bars shown in both Figures 10 and 13 were obtained by choosing the extreme possible locations of the zero-refractivity level. The un-

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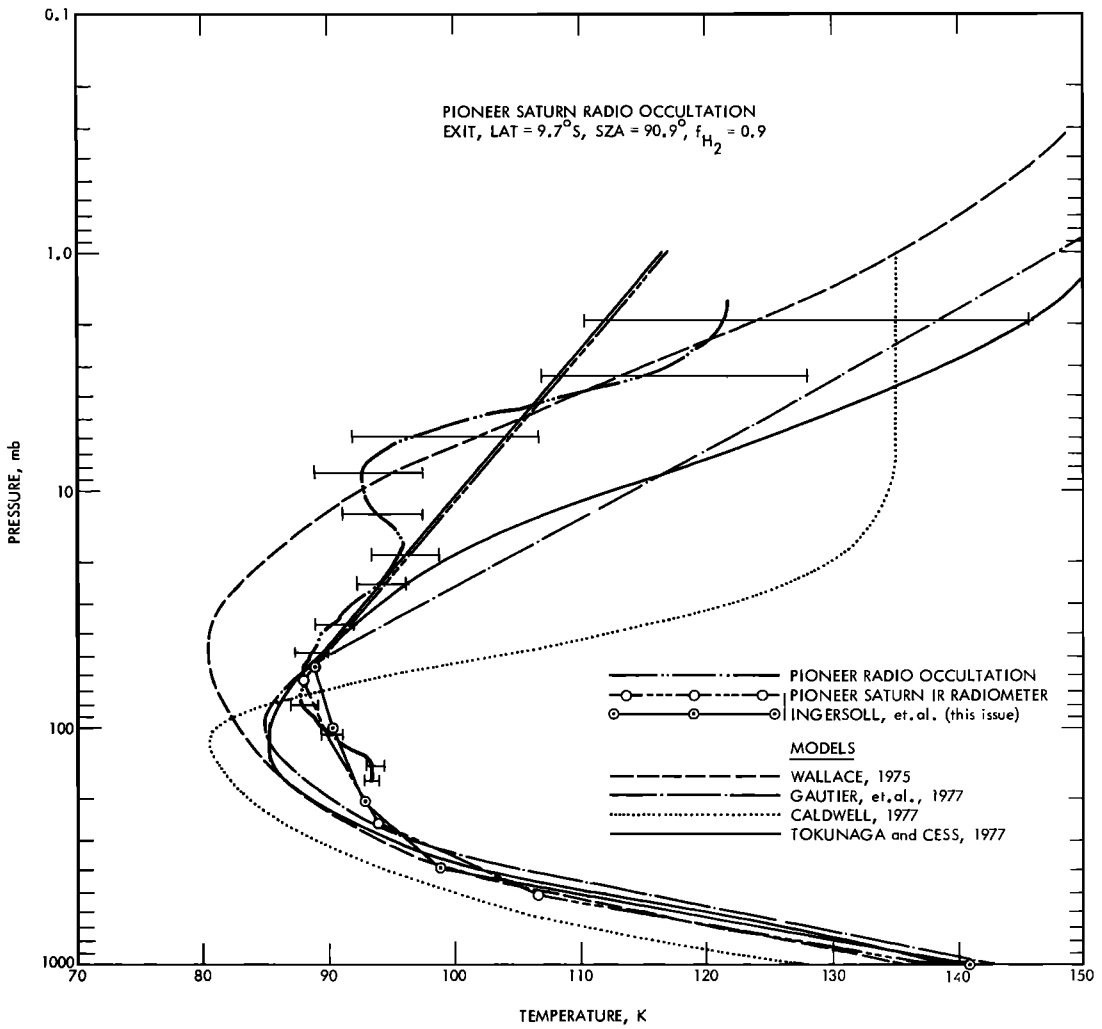


Fig. 13. Occultation temperature profile plotted against pressure and compared with infrared radiometer–retrieved temperature structure and several models derived from earth-based observations.

certainty limits for pressure and temperature shown in Table 1 were also obtained in this manner.

The temperature profile in Figure 10 depicts what is thought to be the temperature minimum in the atmosphere of Saturn. The main minimum is at a radius of about 60,344 km, where the temperature is about 88 K; however, there is a very pronounced minimum of 92.7 K about 80 km higher, and the behavior of the temperature profile at the bottom of the measurement indicates the presence of a third minimum, possibly about 50 km below the main one. Explanations for the causes of these multiple temperature minima are presently under investigation, but they could possibly be artifacts introduced by the spacecraft auxiliary oscillator.

The pressure and number density profiles are shown, in Figures 11*a* and 11*b*, respectively. Since the pressure and number density profiles are only very slightly affected by the choice of the refractivity zero level, the error bars would be less than the width of the lines on the graph. As a reference for the equatorial radius of Saturn the level of 100-mbar pressure is at about 60,332 km, and the level of a number density of 1×10^{19} cm^{-3} is at about 60,328 km. The uncertainty of these radii is a direct reflection of the uncertainty in the spacecraft orbit. Although the formal covariance coefficients of the orbit solutions are very small, there are formidable problems in fitting an orbital state vector, along with satellite masses and other parameters, to data that are heavily perturbed by solar wind fluctuations. Therefore it is quite possible for these radii to be uncertain by up to 30 km. (Pioneer Saturn Navigation Team, private communication, 1980).

One of the most interesting uses that these results can be put to is to attempt to estimate the fraction of helium (the percentage of helium by number, where $fH_2 + fHe = 1.0$) in the Saturn atmosphere by comparing the temperature structure derived from our experiment with those retrieved from the infrared radiometer (IR) measurements [*Orton and Ingersoll,* this issue]. Such an estimate is possible because the IR retrievals are quite insensitive to the helium fraction, and in contrast, the occultation profiles are quite sensitive to the composition assumed, with each percentage point change in the Helium fraction producing a change in temperature of about 0.8 K. In this manner the helium fraction assumed in the computation of the occultation temperature profiles can be varied until a satisfactory fit with the IR temperature retrievals is obtained. However, as shown in Figure 12*a*, the location of the temperature minimum in the IR retrievals is sensitive to the assumption of the steepness of the overlying thermal structure. Thus for a steepness of 32 K/decade it best fits the occultation profile for a hydrogen fraction of 0.94, or 6% helium; for 23 K/

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decade the best fit is for 8% helium; and for 14 K/decade the best fit is for a helium fraction of 10%. The steepness of the overlying structure is not, however, a free parameter. The most reasonable assumption to make is that the slope of the overlying structure should match the slope of the occultation profile at least to a level of about 10 mb, beyond which the sensitivity of the infrared instrument is low. Figure 12*b* shows two such retrievals for a steepness of 16 K/decade corresponding to the average steepness of the occultation profile, with three occultations profiles superimposed for helium fractions of 8, 10, and 12%. It is evident that the best fit is obtained for a helium fraction of 10%. The same conclusion was obtained by the IR experimenters through a totally different method of evaluation of the occultation profiles, involving the computation of the infrared radiance resulting from occultation temperature profiles derived with different values of the helium fraction. By assuming a 3 K uncertainity in the IR temperatures [*Orton and Ingersoll,* this issue] and about a 1 K uncertainty inherent in the occultation profiles, a very conservative uncertainty bound of ±4% is obtained. Therefore the helium fraction by number in the outer atmosphere of Saturn is about 10 ± 4%. This is in substantial agreement with the helium fraction in the Jupiter atmosphere as determined by the Voyager infrared instrument [*Gautier, et al.,* 1981]. The uncertainties are too large to attach a great amount of significance to this finding, but a degree of depletion of helium is suggested by results based on the Saturn gravity field [*Hubbard et al.,* this issue].

For comparison purposes, Figure 13 displays the occultation temperature profile plotted against pressure along with the temperature structure retrieved from the infrared radiometer data [*Orton and Ingersoll,* this issue] as well as models derived from earth-based observations in the infrared [*Wallace,* 1975; *Gautier et al.,* 1977; *Caldwell,* 1977; *Tokunaga and Cess,* 1977]. At the temperature minimum the occultation profile fits the models of *Gautier et al.* [1977] and *Tokunaga and Cess* [1977] reasonably well. However, the upper minimum at about 10-mbar pressure level deviates somewhat from the models. Likewise, the bottom portion of the main minimum in the occultation profile appears to show a temperature increase markedly greater than those displayed by the models or by the IR temperature structure. However, the very bottom portion of the occultation profile displays another temperature decrease after which the signal, unfortunately, disappears. It is quite likely that this temperature decrease marks the beginning of a third temperature minimum, below which the actual temperature structure probably joins the convective region of the atmosphere at the pressure level suggested by the models and by the IR temperature structure. However, the confirmation of this conjecture will have to await the arrival at Saturn of Voyager 1.

It has been mentioned earlier that the entry data were affected by an anomaly which caused large and erratic frequency and amplitude fluctuations immediately below the main peak of the ionosphere. The appearance of this anomaly is shown in Figure 14. Figure 14*a* shows the residuals obtained from the entry open loop data, and Figure 14*b* shows the magnified view of the region affected by the anomaly. For comparison, it must be recalled that the maximum frequency excursion caused by the main peak of the ionosphere was only 0.4 Hz, while the fluctuations in the anomalous region exceed 100 Hz. It is certain that these fluctuations are not directly caused by any radio propagation phenomenon but are probably the result of erratic episodes of loss of lock by the spacecraft transponder. During this time the frequency transmitted to the spacecraft included a frequency rate to aid the spacecraft transponder in the tracking of the uplink signal. At the time predicted to correspond to the region of the top of the atmosphere of Saturn, the frequency rate term was practically reversed in order to allow the uplink to remain in lock as long as possible as the signal penetrated the atmosphere. However, since the predicted times were based on a preencounter orbit solution, there was about a 36-s error between actual and predicted times of occultation events. It is highly probable that the anomalous behavior of the frequency was caused by the inability of the spacecraft transponder to maintain lock when the uplink frequency rate was suddenly reversed in the presence of fluctuations produced by the ionosphere of Saturn and that this intermittent loss of lock continued until the radio

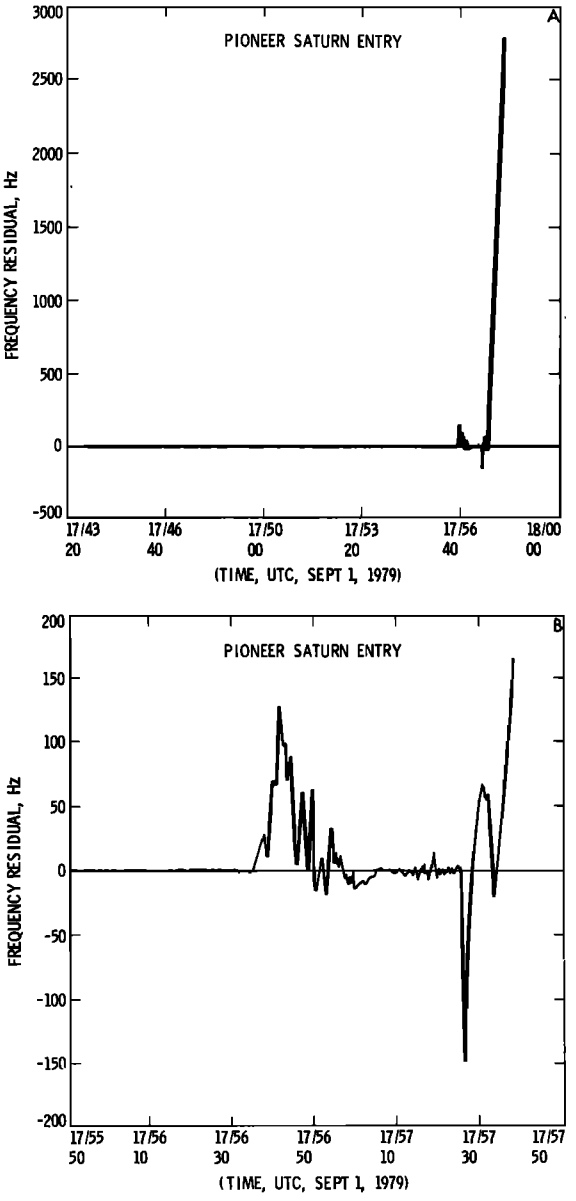


Fig. 14. (*a*) Frequency residuals computed from open loop data taken during entry into occultation showing anomalous fluctuations in frequency which occurred between the bottom of the ionosphere and the top of the neutral atmosphere. (*b*) An expanded view of the frequency fluctuations during the anomalous period.

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link reached the top of the sensible atmosphere of Saturn. If this explanation is correct, then it may very well be possible that information can be retrieved from the portion of the entry data taken within the neutral atmosphere.

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REFERENCES

Anderson, J. D., G. W. Null, E. D. Biller, S. K. Wong, W. B. Hubbard, and J. J. McFarlane, Pioneer Saturn celestial mechanics experiment, *Science, 207,* 449, 1980.

Berman, A., and R. Ramos, Pioneer Venus occultation radio science data generation, *IEEE Trans. Geosci. Remote Sensing, GE-18,* 11, 1980.

Brinkman, A. W., and J. McGregor, The effect of the ring system on the solar radiation reaching the top of Saturn's atmosphere: Direct radiation, *Icarus, 38,* 479, 1979.

Caldwell, J., The atmosphere of Saturn: An infrared perspective, *Icarus, 30,* 493–510, 1977.

Eshleman, V. R., G. L. Tyler, G. E. Wood, G. F. Lindal, J. D. Anderson, G. S. Levy, and T. A. Croft, Radio science with Voyager 1 at Jupiter: Preliminary profiles of the atmosphere and ionosphere, *Science, 204,* 987, 1979.

Fjeldbo, G., and V. R. Eshleman, The atmosphere of Mars analyzed by integral inversion of Mariner 4 occultation data, *Planet. Space Sci., 16,* 1035, 1968.

Gautier, D., A. Lacombe, and I. Reva, Saturn: Its thermal profile from infrared measurements, *Astron. Astrophys., 61,* 147, 1977.

Gautier, D., B. Conrath, M. Flaisar, R. Hanel, V. Kunde, A. Chedin, and N. Scott, The helium abundance of Jupiter from the Voyager infrared investigation, *J. Geophys. Res., 86,* in press, 1981.

Gehrels, T., Bimodality and the formation of Saturn's ring particles, *J. Geophys. Res.,* this issue.

Hubbard, W. B., J. J. MacFarlane, J. D. Anderson, G. W. Null, and E. D. Biller, Interior structure of Saturn inferred from Pioneer 11 gravity data, *J. Geophys. Res.,* this issue.

Kliore, A. J., Current methods of radio occultation data inversion, *NASA Tech. Memo X-62,* 150, 3–2, 1972.

Kliore, A. J., and P. M. Woiceshyn, Temperature in the atmosphere of Jupiter from Pioneer 10 and 11 radio occultations, in *Jupiter,* edited by T. Gehrels, University of Arizona Press, Tucson, Ariz., 1976.

Kliore, A. J., G. F. Lindal, I. R. Patel, D. V. Sweetnam, H. B. Hotz, and T. R. McDonough, Vertical structure of the ionosphere and upper neutral atmosphere of Saturn from the Pioneer radio occultation, *Science, 207,* 446, 1980.

Orton, G. S., and A. P. Ingersoll, Saturn's atmospheric structure and heat budget, *J. Geophys. Res.,* this issue.

Rishbeth, H., S. Ganguly, and J. C. G. Walker, Field-aligned and field-perpendicular velocities in the ionospheric *F*2-layer, *J. Atmos. Terr. Phys., 40,* 767, 1978.

Tokunaga, A., and R. D. Cess, A model for the temperature inversion within the atmosphere of Saturn, *Icarus, 32,* 321, 1977.

Waite, J. H., Jr., S. K. Atreya, and A. F. Nagy, The ionosphere of Saturn: Predictions for Pioneer 11, *Geophys. Res. Lett., 6,* 723, 1979.

Wallace, L., On the thermal structure of Uranus, *Icarus,* 25, 538, 1975.

Wallace, L., and D. M. Hunten, The Lyman-alpha albedo of Jupiter, *Astrophys. J., 182,* 1013, 1973.

Weiser, H., R. C. Vitz, and H. W. Moos, Detection of Lyman α emission from the Saturnian disk and from the ring system, *Science, 197,* 755, 1977.

Woo, R., W. Kendall, A. Ishimaru, and R. Berwin, Effects of turbulence in the atmosphere of Venus on Pioneer Venus radio—Phase I, *JPL Tech. Memo 33-644,* Jet Propul. Lab., Pasadena, Calif., June 1973.

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